

6.2 Fundamental particles

The work here concerns the structure and binding of atoms and nuclei and the nature of fundamental particles. The practical implications of radioactivity are considered, introducing the idea of risk.

The first section (**6.2.1**) considers scattering experiments as a source of evidence about the structure of atoms and nucleons. Ideas from earlier in the module are used to consider particle paths in magnetic and electric fields in the context of particle accelerators. Evidence for discrete energy levels leads on to a crude model of the atom as a particle

in a box. This section gives more opportunities to discuss the development of models in physics and the international cooperation needed to fund large experiments (HSW11).

The second section (**6.2.2**) sees changes in nuclear binding energy per nucleon as driving different types of decay. This leads to a consideration of nuclear power generation. The biological effects of ionising radiation are also considered, giving more opportunity to consider issues of ethics, decision making and the risks and benefits of technology (HSW9, HSW10, HSW12).